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IT403 - Mobile computing

Class Test

3/11/26

(1) Given,

Hexagonal cellular network = 1200 km^2

Cell radius = 1.8 km^2 km

Available spectrum = 48 MHz & full duplex = 200 kHz

Frequency reuse clusters N = 7, N = 4.

Reduction = 15 %.

(i) Area of the Hexagonal cellular network

$$A = 2.598 \times (R)^2$$

$$= 2.598 \times (1.8)^2$$

$$= 2.598 \times 3.24$$

$$A = 8.41$$

Area of Hexagonal cellular network = 8.41

(ii) no of ~~cells~~ total cells in channel

$$N = \frac{\text{Available spectrum}}{\text{~~Freq~~ of duplex channel}} = \frac{48 \times 1000}{200} = 240$$

$N = 240$

It reduces by 15 %

$$= 240 \times 15/100 = 204$$

(iii) Calculate theoritical and effective value

for N = 7

Theoritical $240/7 = 34.28$

effective value

$$204/7 = 29.142$$

For N = 4,

Theoritical value

$$= 240/4 = 60$$

Effective value

$$= 204/4 = 51$$

iv) total available channels:

$$\text{Total available channels} = \frac{\text{cellular network}}{\text{Area of hexagonal}}$$

$$= \frac{1200}{8.41} = 142.68$$

total available channels $\simeq 143$.

v) Cochannel Reuse distance:

$$D/R = \sqrt{3N}$$

$$D = R\sqrt{3N}$$

$$Q = D/R = \sqrt{3N}$$

For N = 7,

$$Q = \sqrt{3} \times 7$$

$$= 12.124$$

For N = 4, $Q = \sqrt{3} \times 4 = 6.928$

Therefore,

For N = 7, provides less channels per each cell and decreases capacity, N = 4 provides more channels with more network capacity.

The min reuse distance is 4.5.

∴ 4 doesn't satisfies the condition.

(2) Given,

PLMNs = 40 min emergency operation

An ambulance = 15 cell boundaries at 90 km/h

2 inter PLMN roaming transitions.

Each service PLMN handoff = 120 ms BSS and 80 ms for NSS

Each inter PLMN = 350 ms for authentication

Roaming registration = 250

Recovery = 400 ms.

(i) The total handoff delay

$$Th = 120 + 80$$

$$= 200$$

$$Th = 200$$

For cell boundaries

$$= 200 \times 15$$

$$= 3000 \text{ ms} \quad \text{or} \quad 3 \text{ s.}$$

(ii) Recovery delay

$$\text{Recovery delay} = 400 \times 3 / 3$$

$$\text{Recovery delay} = 400$$

(iii) roaming related delay

$$T_{roam} = 350 + 250$$

$$= 600$$

for 2 inter PLMN

$$= 600 \times 2$$

$$= 1200$$

iv) Overall communication delay.

overall communication delay

$= T_h + T_R + T_{Roam}/3$

$= 3000 + 400 + 1200/3$

$= 5.4$

v) potential patient data.

Data loss = Data rate × Interruption

$= .512 \times 3$

Data loss = 1536

Convert into KB

$1536/8 = 192$ bits

Potential patient data = 192bits.

crosses the Interruption value.

The min of Interruption is 3, but the actual Interruption is 5.4.

So, it doesn't satisfy the condition

(3) Given,

i) GSM operator.

BSC manages = 12 Base transceiver stations (BTS)

each with 3 sectors

4 transceivers = 4

Each TRx Provides

TDMA = 8.

Timeslots = 2

Each sector serves = 1200 subscribers.

each making = 1.2 calls per hour

Avg call holding time = 90 s

each traffic channel requires = 16 Kbps

E1 link = 2.048 Mbps.

i) Determine the minimum number of TRXs required per sector.

$$C = (TRx \times 8) - 2$$

$$C = (4 - 8) - 2$$

$$C = 30$$

ii) TRXs required per sector connected to BTS

$$= 30 \times 3$$

$$= 90$$

iii) Connected to BCC

$$= 90 \times 12$$

$$= 1080$$

iv) Base station infrastructure

each sector.

$$= \frac{1200 \times 1.2 \times 60}{90} = \frac{86400}{90}$$

$$= 960/10 = 96$$

Channel utilization

$= 96 \times 2.048$

$= 196.608\ \%$

channel utilization = 196.9%

The max utilization is 80%

the actual utilization cross 80%, it becomes 196.9%.

So it doesn't satisfy the condition

(ii) one way spectrum = 10 MHz

FDMA = 30 kHz

Frequency reuse cluster size = 7

TDMA users = 200 kHz

8 users per carrier

N = 4

CDMA = 1.25 MHz.

chip rate = 1.2288 Mcps

voice data rate = 9.6,

voice activity factor = 0.4.

frequency reuse efficiency = 0.6

(i) total channels

total channels = FDMA + TDMA

$= 30 + 200$

$= 230$

Total channels = 230.

(ii) channels per cell for FDMA and TDMA

For FDMA,

N = 7,

FDMA = 230/7

= 32.85

For TDMA,

N = 4

= 230/4 = 57.5

Processing gain and users per carrier for comg

32.85 + 57.5

= 90.35

total users per each

= 90.35 - 15

= 75.35

The utilization is 75.35

The utilization is 80%.

So it satisfies the condition //

(4) Given,

medical imaging files = 240

Size = 18 MB

Theoritical data rate = 20 Kbps

Reduces capacity = 18%

effective throughput = 85%

Packet retransmission = 12%

(i) Analyse GPRS and UMTS by calculating aggregate data

each time slots × Data rate

$= 4 \times 20$

$= 80$

Aggregate data rate = 80

(ii) effective GPRS throughput

effective theoughput = 85%

$= 85/100 \times 240$

$= 204$

effective GPRS theoughput = 204

(iii) total transmitted data volume

total transmitted data volume

= 12% packet retransmission rate each lasts 2.5 sec with 8 mobilities

$= 12 \times 8$

$= 96$

total transmitted data volume = 96

includes interruptions 8 mobilities lasts

$= 96/2.5$

$= 38.4$

total transmitted datavolume including interuptions = 38.4

(iv) Percentage reduction in transfer time achieved by UMTS

= effective · 38.4 + 2.5 = 40.09%

GPRS doesn't meet the 12 hour emergency data delivery deadline, by considering network performance, multiple access techniques and healthcare applications.

(5) Given,

- Software update = 120 GB
  mobile devices = 2000
  GSM = 9.6 kbps per connection
  GPRS effective throughput = 64 kbps with 10% additional info
  5G provides = 25 Mbps per device with 4% protocol overhead
  only 40% transfer data over 5G.

Assume,

1GB = 1000 MB and 1MB = 8 megabits.

i) Data volume per device.

For GSM = 9.6 kbps

$= 9.6 / 8 = 1.2$

For GPRS = 64 kbps + 10%

$= 64/8 + 10/100 = 8 + 1/10 = 8.1$

For 5G = 25 Mbps + 4%

$= 3.125 + 0.25 = 3.165$

ii) Individual transfer times

GSM = $1.2 \times 120 = 144$

GPRS = $8.1 \times 120 = 972$

5G = 3.16 x 120

= 379.2

(iii) Total comparision times per all devices

= Gsm + GPRS + 5G

= 1.2 + 8.1 + 3.16

= 12.46

total comparision times per all devices = 12.46

(iv) Percentage reduction in completion times

Percentage reduction

$= 12.46 \times \left(\frac{40}{100}\right)$

= 12.46 x 0.4

= 4.984

Percentage reduces by 4.984 times.

By comparing 5G and GPRS

5G reduce by 40%.

GPRS = 4.98%

So, 5G reduces nearly 38.02% comparing with GPRS.

By comparing suitability of each technology by considering throughput, transmission overhead, multiple access techniques, user capacity, network evolution. 5G reduces more than GPRS and GSS //